loyal domestic type, but who in reality is concealing a strong tendency towards desertion and unfaithfulness, could have a great advantage. As long as his deserted former wives have any chance of bringing up some of the children, the philanderer stands to pass on more genes than a rival male who is an honest husband and father. Genes for effective deception by males will tend to be favoured in the gene pool.

Conversely, natural selection will tend to favour females who become good at seeing through such deception. One way they can do this is to play especially hard to get when they are courted by a new male, but in successive breeding seasons to be increasingly ready to accept quickly the advances of last year's mate. This will automatically penalize young males embarking on their first breeding season, whether they are deceivers or not. The brood of naïve first year females would tend to contain a relatively high proportion of genes from unfaithful fathers, but faithful fathers have the advantage in the second and subsequent years of a mother's life, for they do not have to go through the same prolonged energy-wasting and time-consuming courtship rituals. If the majority of individuals in a population are the children of experienced rather than naïve mothers—a reasonable assumption in any long-lived species—genes for honest, good fatherhood will come to prevail in the gene pool.

For simplicity, I have talked as though a male were either purely honest or thoroughly deceitful. In reality it is more probable that all males, indeed all individuals, are a little bit deceitful, in that they are programmed to take advantage of opportunities to exploit their mates. Natural selection, by sharpening up the ability of each partner to detect dishonesty in the other, has kept large-scale deceit down to a fairly low level. Males have more to gain from dishonesty than females, and we must expect that, even in those species where males show considerable parental altruism, they will usually tend to do a bit less work than the females, and to be a bit more ready to abscond. In birds and mammals this is certainly normally the case.

There are species, however, in which the male actually does more work in caring for the children than the female does. Among birds and mammals these cases of paternal devotion are exceptionally rare, but they are common among fish. Why?* This is a challenge for the selfish gene theory which has puzzled me for a long time. An ingenious solution was recently suggested to me in a tutorial by Miss T. R. Carlisle. She